Center-Boundary Dual Attention for Oriented Object Detection in Remote Sensing Images

Shuai Liu, Lu Zhang, Huchuan Lu, *Senior Member, IEEE*, and You He

***Abstract*— Recently, anchor-free object detectors have shown promising performance in oriented object detection on remote sensing images. However, the objects in remote sensing images always have large variations in arbitrary orientations, sizes, and aspect ratios, which makes the existing anchor-free methods hard to obtain satisfactory results. In this article, we propose a novel anchor-free detector, center-boundary dual attention (CBDA) network (CBDA-Net), for fast and accurate oriented object detection on remote sensing images. In CBDA-Net, we construct a CBDA module, which utilizes a dual attention mechanism to extract attention features on the center and boundary regions of objects. The CBDA module can learn more essential features for rotating objects and reduce the interference from complex background. Besides, to resolve the influence of object aspect ratio on angle errors, we propose an aspect ratio weighted angle loss (arwLoss), where diffident penalties are assigned on the angle loss based on the aspect ratios of objects. This loss construction is effective in improving the detection accuracy of oriented objects, especially for slender objects. We conduct extensive experiments on two publish benchmarks, i.e., DOTA and HRSC2016. The experimental results demonstrate that our CBDA-Net achieves favorable performance against other anchor-free state of the arts with a real-time speed of 50 FPS.**

***Index Terms*— Anchor-free, dual attention, oriented object detection, remote sensing images.**

I. Introduction

OBJECT detection plays an very important role in the field of remote sensing, which has been applied to many scenarios such as urban planning [1], [2], traffic monitoring [3], and precision agriculture [4]. In remote sensing images, the objects always distribute close to each other with various arbitrary orientations. Using a horizontal bounding box to detect objects in the remote sensing images might lead to ambiguous identification. It is more reasonable to exploit oriented bounding boxes to capture the object characteristics of remote sensing images [5]–[7].

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With the development of aerospace remote sensing technology, it becomes easier to obtain large-scale remote sensing data. How to detect the oriented objects from huge remote sensing image data with high accuracy and fast speed has become a long-term issue in the literature [8], [9]. Recently, with the success of deep learning in visual recognition tasks [10]–[12], convolutional neural networks (CNNs) have been applied to oriented object detection and shown favorable performance [13]–[15].

Existing oriented object detection methods [16]–[18] usually leverage the advances from general object detection approaches [19]–[22]. The common modification is to predict an extra rotation angle on the bounding box and conduct specialized designs to make the model better fix the oriented objects. For instance, Ding *et al.* [23] designed a new network structure to efficiently capture the feature representations of oriented objects. Zhu *et al.* [24] focused on constructing a loss function to solve the periodic problem of rotation angle. These methods have achieved promising results and improve the accuracy of oriented detection in remote sensing images.

However, most of the existing oriented detection methods are built on anchor boxes, in which thousands of anchor boxes are predefined for subsequent box recognition and regression. Such operation brings large hyperparameters and computation cost and thus decreases the inference speed in real-world applications.

On the contrary, anchor-free approaches [25]–[27] do not contain the anchor generation process, where objects are detected using key points. These anchor-free methods could achieve comparable performance with anchor-based approaches while largely increasing the processing speed. Inspired by this, researchers begin to explore the anchor-free oriented object detection methods for remote sensing images. However, directly applying the anchor-free framework may not effectively handle the complicated object distributions in remote sensing images. The problems remain in two aspects: first, localizing objects using key points might be influenced by ambiguous areas from other objects or complicated background (see Fig. 1), and second, anchor-free methods usually optimize the rotation angle as an independent variable without any anchor constraint [28]. The angle error of oriented object would largely influence the accuracy of IoU, especially on objects with higher aspect ratios (see Fig. 2). It is more reasonable to take the effect of aspect ratio into consideration during angle loss construction.

In this article, we propose a novel anchor-free architecture, center-boundary dual attention (CBDA) network (CBDA-Net), for oriented object detection in remote sensing images. First,

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(a) (b)

Fig. 1. Key point ambiguity in anchor-free object detectors. (a) Ambiguous with background. There are many areas similar to the center point of the roundabout in the background as shown by the yellow dots. (b) Ambiguous with other categories. The ground track field always has the same center point as the soccer ball field. If we use the center point of an object to locate and classify, it is possible to get ambiguous predictions and it is also harmful to the network.

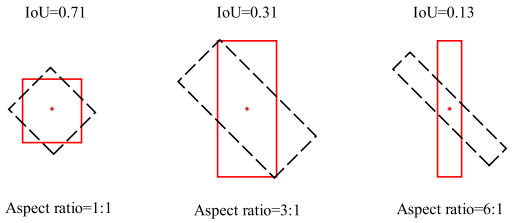


Fig. 2. Influence of angle errors under different aspect ratios on IoU. Object with a higher aspect ratio has a lower IoU under the same angle error. If the angle of the object is regressed directly without considering the influence of the aspect ratio, it is detrimental to the detection results.

to reduce the influence from other objects and complicated background on key point localization, we design a CBDA module. The CBDA module builds a spatial attention mechanism on both center and boundary regions of the oriented objects. Then, the two attention features are adaptively merged to formulate the object classification and oriented bounding box regression. The CBDA module leverages the merits of various region attentions, which could help to generate more accurate key point localization for objects with different aspect ratios. Second, considering that the objects with various aspect ratios have different sensitivities on angle errors, we propose a new aspect ratio weighted angle loss function (arwLoss). In arwLoss, the angle loss weight is automatically assigned according to the aspect ratio of the object. Specifically, objects with larger aspect ratios will obtain higher penalties and vice versa. To investigate the effectiveness of our proposed CBDA-Net, we conduct thorough experiments, including overall comparison and ablation study on two benchmark data sets [29], [30]. The results verify that our method performs favorably against state-of-the-art anchor-free methods.

The main contributions of this article are summarized as follows.

1) We propose a CBDA module to adaptively capture the features from the center and boundary regions of objects, which could improve the object localization accuracy and suppress the interference from complex background.
2) We propose an angle regression loss function to fix the sensitivity of object aspect ratio on angle error. The new loss construction could largely improve the accuracy of oriented object detection, especially for slender objects.
3) We conduct extensive experiments on two public data sets to demonstrate that our proposed method could achieve superior results against state-of-the-art anchor-free approaches.

II. Related Work

A. Oriented Object Detection

Oriented object detection represents the objects by rotating bounding boxes, which is more suitable for special scene detection tasks where objects are randomly distributed, such as scene text detection [31]–[33] and remote sensing images object detection [16]–[18].

Oriented object detection methods usually follow the general object detection paradigm [19], [21], [22], [34], [35] and introduce a rotation angle on the horizontal bounding box. Recently, many methods are designed in the literature to better capture the characters of oriented objects. For example, to better extract the features of the oriented object, Ma *et al.* [31] drawn on the framework of R-CNN and proposed an oriented object detection network, which regresses the oriented bounding box from the rotation region features of interest. Cheng *et al.* [36], [37] proposed rotation-invariant convolutional neural networks to resolve the ambiguous features of rotated objects in remote sensing images. Yang *et al.* [38] designed a feature refinement module on one-stage detector network, which turns the general object detector into an oriented object detector and further improves the detection accuracy. In addition to effective oriented object feature extraction, some methods explore the effective representation of oriented bounding box. For instance, Xu *et al.* [39] described a multioriented object using gliding vertex on the horizontal bounding box, which avoids the use of rotation angle. Zhu *et al.* [24] adopted the 2-D vectors to represent the rotation angle and proposed a length independent IoU calculation method that better detecting for long objects. Moreover, some scholars study the problem that the angle is difficult to regress. For instance, Yang and Yan [40] designed a circular smooth label to transform the angle regression task into a classification task, which avoids the periodicity problem of angle regression.

Anchor-free detectors for oriented objects have achieved increasing attention due to the simple structure without anchor predefinition. Xiao *et al.* [41] described the oriented object by predicting the principal axis and length vertical on the principal axis in anchor-free style. Chen *et al.* [42] proposed an IoU loss, which considers the influence of the large aspect ratio and strengthens the connection between the angle error and IoU. Pan *et al.* [28] designed a dynamic refinement network, which uses a feature selection module to extract oriented object features according to object shape and orientation. These anchor-free object detection methods mostly follow the key points detection paradigm, locate and classify by predicting the key points of the object, and directly regress the geometric information of the object bounding box.

B. Anchor-Free Object Detection

The anchor-free methods remove the process of anchor redefinition, which belongs to bottom-up detection methods and can directly regress the object bounding box through a series of convolution calculations. The overall framework of

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the anchor-free detection method is simple and has become a research hotspot in the field of object detection recently. The anchor-free method originated from Densebox [43], which proves the effectiveness of the object detection algorithm without anchors. The anchor-free methods can be divided into two families: point-grouping methods and point-vector methods.

Point-grouping methods represent the object by a group of key points and then convert these key points into bounding boxes using postprocessing operations. For example, Law and Deng [44] proposed a CornerNet in key points detection paradigm, which utilizes two key points (upper left corner and lower right corner) to represent the object and restores the object bounding box by merging the two points. Zhou *et al.* [45] determined the location and category of object by detecting four extreme points (top, bottom, left, and right) of the object. Yang *et al.* [46] developed the representation of key points by multiple points and convert these sample points into final bounding box by a predefined converting function.

Point-vector methods utilize one key point to represent the location of one object and a vector to represent the geometric information of bounding box. For example, Zhou *et al.* [25] drawn on the key point detection paradigm and proposed an anchor-free object detection method, which takes the center point of object as key point and directly regresses the size of bounding box from feature maps directly. Tian *et al.* [26] proposed a fully convolutional one-stage object detection (FCOS) method, which predicts objects in pixelwise and introduces a centerness branch to constrain the bounding boxes close to object center. Kong *et al.* [27] proposed a new sampling paradigm, in which the positive samples are concentrated in the center area of object to better distinguish them from the negative samples.

The success of anchor-free detection methods lies in the correct detection of key points. Our method belongs to anchor-free methods, and however, we do not only extract the key point information simply but also focus on the key regions (center regions and boundary regions) features of the oriented objects in the process of feature extraction. Our method enhances the representation of objects, especially for the characteristics of oriented objects.

C. Attention Mechanism

Attention mechanism can help the network focus on the information of interest while ignoring irrelevant information, which has a wide range of applications in computer vision tasks, such as classification [47], [48], segmentation [49], and detection [50]. The widely used attention mechanisms include spatial attention mechanism, channel attention mechanism, and spatial channel hybrid attention mechanism. In general, the attention mechanism learns the weights of feature map and applies them on features to strengthen the important information. For instance, Hu *et al.* [47] proposed a squeeze-and-excitation attention network, which learns different weights for different feature channels and completes the correct classification. Fu *et al.* [49] adopted spatial attention to learn the correlation in spatial positions and utilized the channel attention mechanism to learn the semantic correlation between different categories and combine the spatial attention features and channel attention features together.

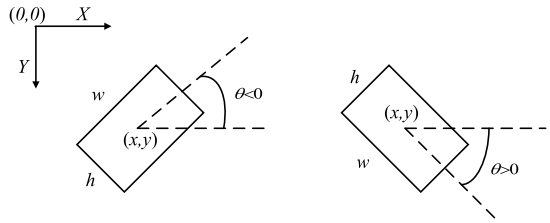


Fig. 3. Details of five-parameter representation of the oriented bounding boxes in our method. (x, y) are the center point coordinates of oriented bounding box, w is the longer side of the bounding box, h is the shorter side of the bounding box, and θ indicates the minimum included angle between w and the positive direction of the x-axis. When the included angle side is above the x-axis, $\theta < 0$; when the included angle side is below the x-axis, $\theta > 0$.

In remote sensing images, some scholars have also applied attention mechanism to object detection and achieved good results. For example, Chen *et al.* [51] combined scale information with spatial and channel attention mechanism to detect small objects effectively. Cui *et al.* [52] proposed a dense attention pyramid network in SAR images for multiscale ship detection. Yang *et al.* [53] introduced a supervised pixel attention network and a channel attention network, which have achieved great performance in densely arranged small object detection. Wang *et al.* [54] proposed a skip-connected encoder–decoder model, which connects information at different scales and learns attention weights in different scale features.

Our proposed CBDA-Net also applies the attention mechanism and can handle multicategory oriented object detection in remote sensing images. Our method not only allows the network to focus on the key regions of objects but also reduces interference and eliminates complex background noise.

III. Methodology

In this section, we first give the problem definition of the oriented bounding box. Then, we introduce the overall framework of our proposed method and CBDA module. Finally, we propose the details of arwLoss.

A. Overall Framework

For a given remote sensing image I, our method aims to detect the objects in C categories and represent their locations using rotated bounding boxes. We use five parameters $(x, y, w, h,$ and $\theta)$ to represent the oriented bounding box, which indicates the center coordinates, width, height, and angle, respectively. To facilitate angle regression and increase the discrimination of rotation angle, we define the rotation angle θ in the range of $[-90°, 90°]$, as shown in Fig. 3.

The overall framework of the proposed CBDA-Net is shown in Fig. 4. The CBDA-Net consists of three submodules, which are backbone network, CBDA module, and oriented object detection head. First, we use the backbone network to extract

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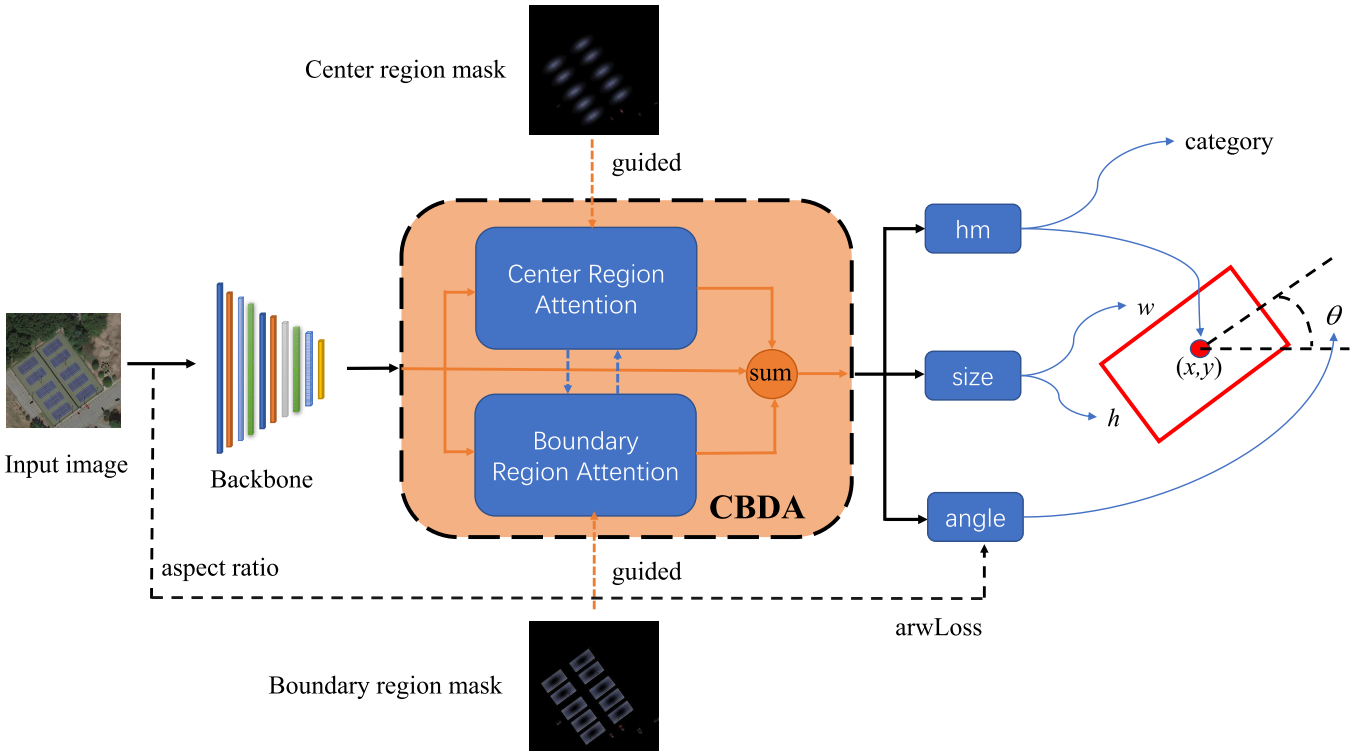


Fig. 4. Framework of our proposed method CBDA-Net. The backbone network is followed by the CBDA module and oriented object detection head. The CBDA module contains dual attention on the center region and the boundary region of the object. The oriented object detection head contains the size and rotation angle of the oriented bounding box as well as the heatmap (hm) of the center point of the object.

features from an input remote sensing image. Then, the CBDA module utilizes a center region mask and a boundary region mask for network attention selection, which could highlight oriented object features and suppress the complex background interference. Then, we obtain fused image feature maps by combining object center features with boundary features. In the oriented object detection head, the rotation angle, size, and the heatmap of the center point are directly regressed from the fused feature maps. In the process of angle regression, we use arwLoss to optimize the rotation angle. Finally, we obtain the center point and category of the object through heatmaps, so as to obtain the final orientated object detection results.

B. Center-Boundary Dual Attention Module

As mentioned in Section II, the background of remote sensing image is complex and the objects have large variations in scale and orientation distribution. In particular, the number of small objects is large and their distributions are irregular. The key points of these small objects are usually ambiguous with background and objects in other categories. Besides, it is difficult to capture the characteristics of the oriented objects using only key point features.

Recent methods Dense RepPoints [55] and BorderDet [56] have demonstrated that the boundary features play a major role when the region features are used to enhance the single-point features. Boundary features can facilitate object localization and capture the size of objects. FoveaBox [27] points out that human beings mainly focus on the central region of objects, which is more adaptable to the shape and spatial distribution of the bounding box. Liu *et al.* [57] proved that the combination of various components can improve the accuracy of detection. From the abovementioned works, we can see that the center region features and the boundary region features are both important in the object feature extraction.

Inspired by these works, we introduce a CBDA module to make good use of the center region features as well as boundary region features for accurate oriented object detection. In the CBDA module, we fully consider the oriented features of objects and use a dual attention mechanism to extract the two region features. In the process of feature extraction, we design center region mask and boundary region mask to assist supervised learning and fuse these two region features by a simply selective operation. This CBDA module can completely extract objects features, especially for the oriented bounding boxes.

1) Architecture: The architecture of the CBDA module is shown in Fig. 5. First, we get the feature $\mathbf{F}$ through backbone network. Then, we use one 1×1 convolution layer to get the center region features $\mathbf{F_C}$ and boundary region features $\mathbf{F_B}$, followed by batch normalization and ReLU activation function in sequence. Next, we use a stack of 3×3 convolution, ReLU activation function, and 1×1 convolution to generate the center region mask $\mathbf{M_C}$. Concurrently, we obtain the center region attention $\mathbf{A_C}$ through convolution operation and the sigmoid activation function. We get the boundary region mask $\mathbf{M_B}$ and boundary region attention $\mathbf{A_B}$ from $\mathbf{F_B}$ in the same way. The shapes of $\mathbf{A_C}$ and $\mathbf{A_B}$ are both $C \times H \times W$, where C is

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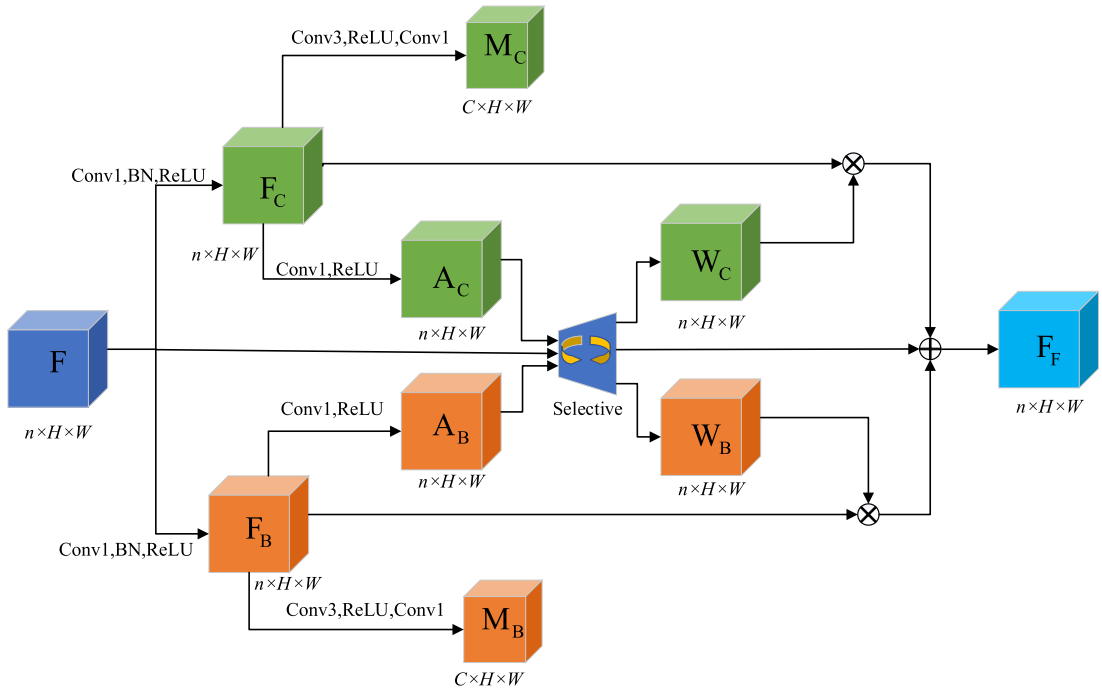


Fig. 5. Framework of CBDA module. CBDA module builds two parallel streams for constructing center attention and boundary attention. Taken the extracted feature map as input, it first generates the center and boundary region features, which are denoted as $\mathbf{F_C}$ and $\mathbf{F_B}$. Then, these two region features are used to produce the center/boundary mask and attention, respectively. The attentions from two regions along with the input feature F are sent to the selective operation to generate center and boundary attention weights, which are further combined with center and boundary region features by multiplication. Finally, the attentioned center and boundary features are integrated with input feature to form the fused feature $\mathbf{F_F}$. conv1 and conv3 represent the convolution of the convolution kernel 1×1 and 3×3, respectively. BN stands for batch normalization and ReLU represents the activation function of the rectified linear unit.

the number of object categories and H and W are the sizes of $\mathbf{F}$, which are 1/4 of the size of input image. We utilize selective operation for the two-region attention and obtain attention weights $\mathbf{W_C}$ and $\mathbf{W_B}$. Finally, we use elementwise multiplication between attention weights and region features to get new center region attention features and boundary region attention features, which are further fused with original features $\mathbf{F}$ together to generate fusing features $\mathbf{F_F}$. The fusion process can be described as

$$\mathbf{F_F} = \mathbf{W_C} \cdot \mathbf{F_C} + \mathbf{W_B} \cdot \mathbf{F_B} + \mathbf{F}. \tag{1}$$

2) Region Mask Representation: In the process of attention learning, we design the center region mask and the boundary region mask for supervised learning. A previous method [58] proposed to use the object mask for supervision to enhance the feature representation. However, it simply produces a binary mask by labeling the pixels inside bounding box as 1. Such binary mask cannot capture the shape of object and includes some noisy background areas around the object. To better represent the shape and position of rotating object, we design two heatmap masks on the center and boundary region to capture more complete information of the object.

In the center region mask representation, more attention should be paid on the center points that we use to locate the objects. When formulating the center region mask, we use a Gaussian function on the center point of the object to produce a heatmap. In this heatmap, the most highlighted region represents the object center and the pixel closer to center is assigned with a score closer to 1. Furthermore, to reflect the rotating shape of the objects, we give different downward speeds from the center points to the object's boundary. The closer to the direction of the rotation angle, the slower the value drops. To achieve this effect, we design a function to generate the center region mask $\mathbf{M_C}$ by a modified Gaussian function, and it is defined as

$$\mathbf{M_C} = e^{\frac{-[\mathrm{Rot}(\theta)(P-P_C)]^2}{R^2 \mathrm{rat}^2}} \tag{2}$$

$$\mathrm{Rot}(\theta) = \begin{bmatrix} -\sin\theta & \cos\theta \\ \cos\theta & \sin\theta \end{bmatrix} \tag{3}$$

$$\mathrm{rat} = w/h \tag{4}$$

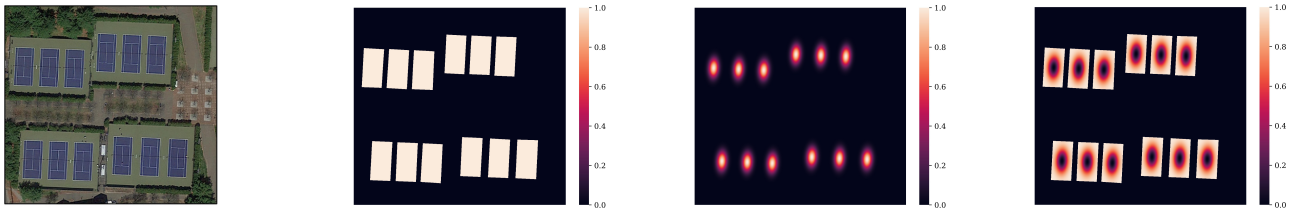
where P is the position of each point on the center region mask, P_c is the center point of the object, and $\mathrm{Rot}(\theta)$ is the rotation matrix, related to the rotation angle θ. rat is the aspect ratio of the object bounding box, and it is always bigger than 1 because we previously defined w to be greater than h. R is the Gaussian kernel function variance and we take 1/3 of h as R in our method.

Similar to the center region mask, the boundary region mask $\mathbf{M_B}$ focuses on the object boundary regions. As shown in Fig. 6(d), the boundary pixels of the object are set to 1, and the values of other pixels are gradually decreasing toward the center of the object. The boundary region mask $\mathbf{M_B}$ can be described as

$$\mathbf{M_B} = \mathbf{M} - \mathbf{M_C} \tag{5}$$

where $\mathbf{M}$ is the traditional binary mask generated by four vertices of the oriented bounding boxes.

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(a) (b) (c) (d)

Fig. 6. Raw image and corresponding different mask representations. (a) Raw image. (b) Mask generated by four vertices of oriented bounding box. (c) Our proposed center region mask. (d) Our proposed boundary region mask.

3) Selective Operation: The selection operation can extract satisfying object features and plays a vital role in the CBDA module. The basis of selection operation is to select the most effective attention for each region. Taking center attention for example, by comparing it with boundary attention in pixel-wise, the positions with larger attention values than boundary attention will be processed by sigmoid-based formalization; otherwise, they will be set to 0. This selection process is conducted in the boundary attention map. The selection process is as follows:

$$\mathbf{W_C} = \begin{cases} \text{Sigmoid}(\mathbf{A_C}) & \text{if } \mathbf{A_C} > \mathbf{A_B} \\ 0 & \text{else} \end{cases} \tag{6}$$

$$\mathbf{W_B} = \begin{cases} \text{Sigmoid}(\mathbf{A_B}) & \text{if } \mathbf{A_B} > \mathbf{A_C} \\ 0 & \text{else} \end{cases} \tag{7}$$

where $\mathbf{W_C}$ is the center region attention weight and $\mathbf{W_B}$ is the boundary region attention weight.

To better illustrate the effectiveness of the CBDA module, we draw the attention heatmaps before and after the CBDA module in Fig. 7. Especially, the input image passes through the backbone network and CBDA module in turn to complete the image feature extraction, and we draw the attention heatmaps before and after entering the CBDA module and analyze the effect of using the CBDA module. We can observe that the CBDA module can effectively suppress background noise interference and focus on the important areas of the targets. For example, it focuses more on the boundary region of tennis court while focusing on both two regional features of ground track field.

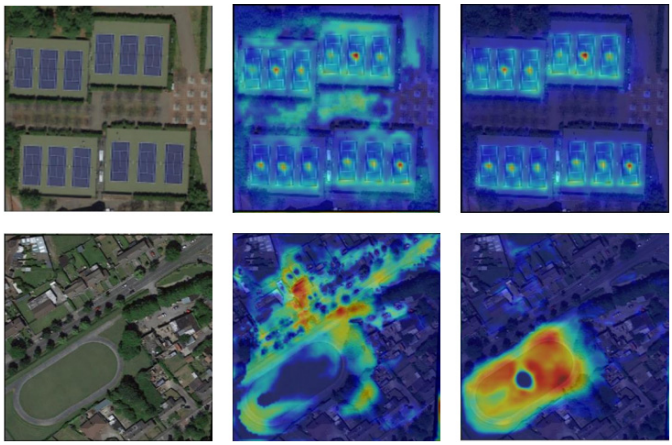


Fig. 7. Visualization of attention heatmaps of CBDA module. (From Left to Right) Input image, attention heatmap without CBDA module, and attention heatmap with CBDA module. CBDA module can focus on representative features of the target while eliminating background interference.

C. Aspect Ratio Weighted Angle Loss

Remote sensing images not only have a complex background but also have a variety of objects, with different scales and aspect ratios. We make statistics on the aspect ratio of each category object in the DOTA training data set and present the results in Fig. 8. In order to express the statistical results more appropriately, we use $1/\text{rat}$ as the ordinate and limit the ordinate between [0, 1]; $1/\text{rat}$ is closer to 0, and the object is potential to be narrower with a larger aspect ratio. It is worth noting that different categories of objects have different aspect ratios in remote sensing images. For example, the aspect ratios of storage-tank and roundabout close to 1, but the ship and large-vehicle's aspect ratios are more than 25. Except for the variation in aspect ratio among different categories, in the same category, the difference in the aspect ratios of objects is relatively large. For example, the maximum aspect ratio is close to 50, and the minimum is only 1 in the harbor category.

In order to explore the influence of angle errors on object detection accuracy under different aspect ratios, we calculate the Intersection over Union (IoU) with different aspect ratios and angle errors and list the results in Fig. 8. In oriented object detection, the IoU between the two object bounding boxes is not only influenced by the rotation angles but also related to the sizes and locations of the two object bounding boxes.

IoU has different sensitivities to angle errors under different aspect ratios. As shown in Fig. 9, it is obvious that different angle errors have different influences on IoU. Under the same aspect ratio, the larger the angle errors are, the smaller the IoU is. At the same time, we can also find that the IoU caused by different aspect ratios is different in the same angle errors. The higher the aspect ratio is, the smaller the IoU is. In other words, under the same angle errors, the narrower and longer the object is, the greater the deviation between the predicted oriented bounding box and ground truth.

In order to eliminate the unbalance influence of the aspect ratios on detection accuracy, we define an angle weighted loss, arwLoss, which considers the effect of aspect ratio on angle regression. Considering that the angle errors under different

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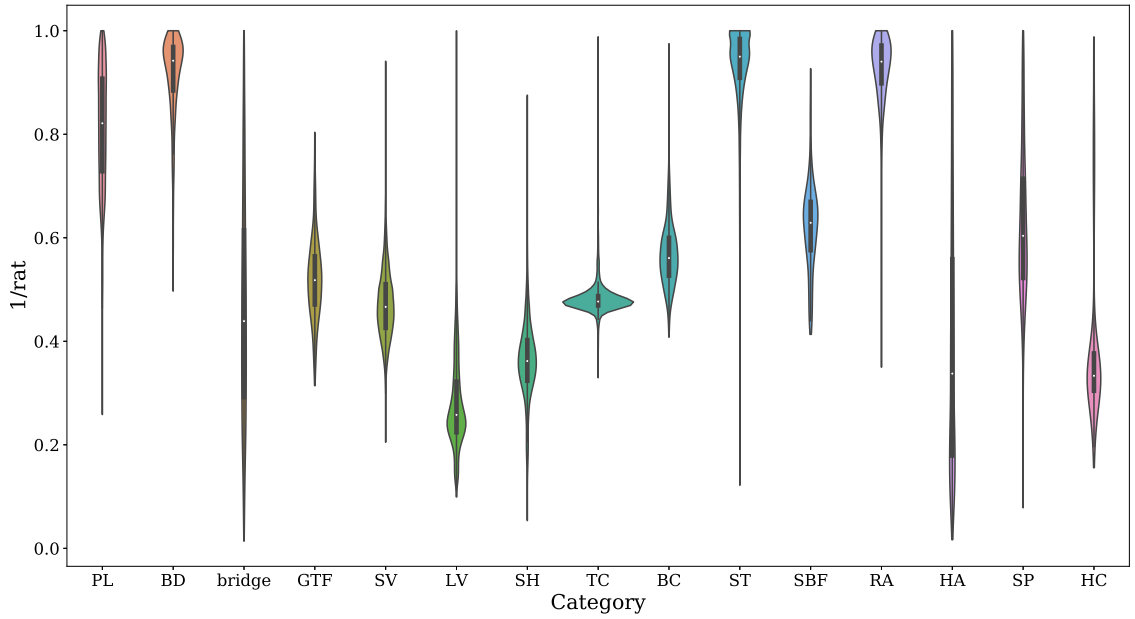


Fig. 8. Distribution of aspect ratios of each category in the DOTA training data set. Aspect ratios are quite different for which objects in the same category or different categories. Plane (PL), baseball diamond (BD), bridge, ground field track (GFT), small vehicle (SV), large vehicle (LV), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC).

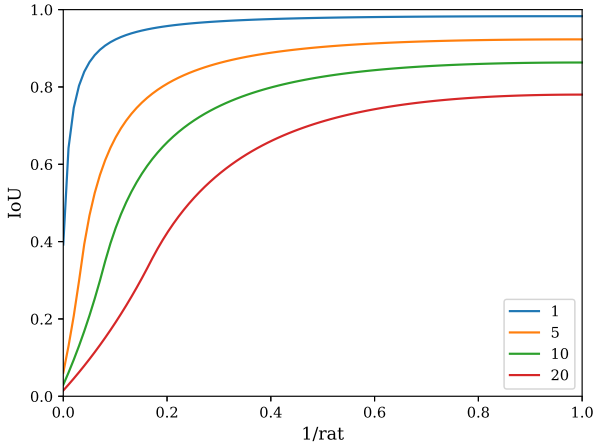


Fig. 9. IoU curves under different angle errors and different aspect ratios. The curves of different colors represent different angle errors, label "1" represents the error of 1°, label "5" represents the error of 5°, and so on. IoU has a different sensitivity to angle errors under different aspect ratios. The higher the aspect ratio, the smaller the IoU under the same angle errors.

aspect ratios have different effects on the accuracy of object detection, we hope that the angle losses under different aspect ratios are different, and the angle regress losses of objects with higher aspect ratios should be larger. The arwLoss is defined as

$$L_{\text{arw}}(\theta_i, \theta_i^*) = \sum_{i=1}^{N} \left\{ (1 + e^{-1/\text{rat}_i}) \left| \theta_i - \theta_i^* \right| \right\} \tag{8}$$

where i is the index of objects in one batch, θ_i is the predicted rotation angle in i target object, θ_i^* is the ground truth of rotation angle in i target object, rat_i is aspect ratio in i object, and N is the number of target objects in one batch.

IV. Experiments

A. Experimental Setup

1) Data Sets: To prove the effectiveness of our method, we conduct a series of experiments on two data sets: DOTA data set and HRSC2016 data set.

DOTA data set is a remote sensing image data set used for object detection. It contains objects in 15 categories and a total of 188 282 instances. Each instance is labeled by the four vertices of the rotating bounding box. The resolution of each image is about $4k \times 4k$. The entire data set is arbitrarily divided into three parts, where one-half of the original images are selected as the training set, 1/6 as the verification set, and 1/3 as the test set [29].

The HRSC2016 data set is used for ship detection, which has 1070 images and 2970 instances. Each instance uses two types of labels: rotating bounding box and horizontal bounding box. The image size ranges from 700 to 1100. The data set is divided into the training set, validation set, and test set, which contain 444, 436, and 188 images, respectively [30].

2) Implementation Details: The image size of the DOTA data set is generally large, and we split the images into 1024 × 1024 pixels subimages with 200 pixels overlap. In order to reduce the negative impact of category imbalance, we have made rotation augmentation for small category objects and enhance the rotation data of 90°, 180°, and 270° in the small category of objects. When using multiscale testing, we scale the test set images by 0.5 times and 1.5 times before splitting the large images, and then, we use NMS to process the test results and the NMS threshold is set to 0.1. We do not process the data of the HRSC2016 data set and only adjust it to an image of 512 × 512 pixels for training and testing.

3) Evaluation Metrics: To compare and measure the performance of the different methods fairly, we adopt mean average

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TABLE I
QUANTITATIVE COMPARISONS BETWEEN OUR METHOD AND OTHER METHODS FOR OBB TASK ON DOTA TEST SET

Method	mAP(%)	PL	BD	Bridge	GFT	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Anchor-based method																
RRPN [31]	61.01	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58
RADet [17]	69.09	79.45	76.99	48.05	65.83	65.46	74.40	68.86	89.70	78.14	74.97	49.92	64.63	66.14	71.58	62.16
RoI-Transformer [23]	69.56	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67
CAD-Net [59]	69.9	87.8	82.4	49.4	73.5	71.1	63.5	76.7	**90.9**	79.2	73.3	48.4	60.9	62.0	67.0	62.2
SCRDet [53]	72.61	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	65.02	66.68	66.25	68.24	65.21
R^3Det [38]	72.81	89.24	80.81	51.11	65.62	70.67	76.03	78.32	90.83	84.89	84.42	65.10	57.18	68.10	68.98	60.88
Gliding Vertex [39]	75.02	89.64	85.00	52.26	77.34	73.01	73.14	**86.82**	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32
APE [24]	75.75	89.96	83.62	53.42	76.03	74.01	**77.16**	79.45	90.83	87.15	84.51	67.72	60.33	74.61	71.84	65.55
CSL [40]	76.17	90.25	85.53	54.64	75.31	70.44	73.51	77.62	90.84	86.15	86.69	**69.60**	68.04	73.83	71.10	**68.93**
HSP [58]	**78.01**	**90.39**	**86.23**	**56.12**	**80.59**	**77.52**	73.26	83.78	90.80	87.19	85.67	69.08	**72.02**	**76,98**	**72.50**	67.96
Anchor-free method																
Axis Learning [41]	65.98	79.53	77.15	38.59	61.15	67.53	70.49	76.30	89.66	79.07	83.53	47.27	61.01	56.28	66.06	36.05
O2-DNet [60]	71.04	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03
DRN [28]	73.23	**89.71**	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	**69.30**	69.63	58.48
Ours	**75.74**	89.17	**85.92**	**50.28**	**65.02**	**77.72**	**82.32**	**87.89**	90.48	**86.47**	**85.90**	**66.85**	**66.48**	67.41	**71.33**	**62.89**

precision (mAP) and frames per second (FPS) to measure the accuracy and efficiency of the detection algorithm. In addition to mAP and FPS, we also use precision–recall (PR) curves and IoU to evaluate the quality of the detection algorithm. These methods are all commonly used in object detection.

The PR curves metric measures the precision and recall of object detection results. The precision is the proportion of true positives in the detection results and the recall is the proportion of true positives in all objects to be detected. The Precision and Recall metrics can be formulated as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

where TP represents the number of positive samples that are correctly identified, FP represents the number of negative samples that are incorrectly identified as positive samples, and FN represents the number of positive samples that are incorrectly identified as negative samples.

The mAP is the mean value of AP values for all categories in the data set. The AP is the average of the Precision value on the PR curves. This article uses the area under the PR curve as AP value.

4) Parameter Settings: During the training phase, we resize the DOTA subimages to 800 $\times$800 pixels and arbitrarily scale the image between 0.6$\times$ and 1.4$\times$. The backbone network is DLA-34 with deformable convolution in our method. We train the network for 140 epochs and the initial learning rate is 1.25e$-$4. The learning rate decays ten times at 90 epochs and 120 epochs. We use multiple loss functions for joint training. The heatmap loss, center region mask, and boundary region mask all use the modified focal loss to supervise. We use the L1 loss function and the proposed arwLoss to supervise the size and rotation angle of the object, respectively. Except for the heatmap loss's weight that is 1, other loss functions' weights are set to 0.1. We use the SGD optimization function for optimization in our experiments with batch size 12, on two GeForce GTX 2080ti GPUs.

During inference, we extract the peaks in the heatmap for each category independently. When the center point score is greater than 0.1, it is considered that the object center point is correctly detected.

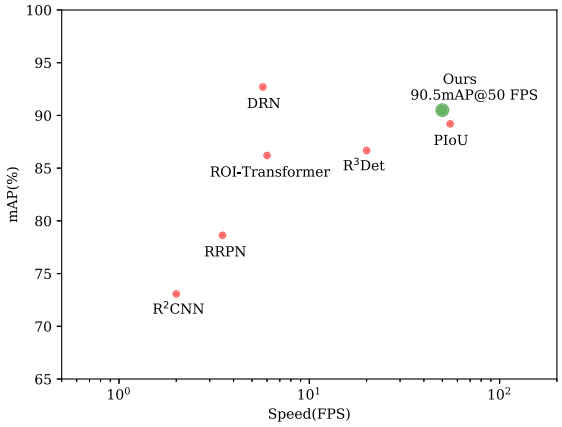


Fig. 10. Performance versus speed on the HRSC2016 data set. As can be seen, our method significantly surpasses competitors in accuracy while running very fast.

B. Comparisons With State of the Arts

1) Results on DOTA Data Set: We compare our method with both anchor-based and anchor-free approaches on the DOTA test set. In comparison experiment, we use multiscale test and submit our prediction results to the official DOTA evaluation server. The comparison performance between our method and other methods is shown in Table I. The corresponding visualization results of the oriented bounding box (OBB) object detection task on the DOTA test set are presented in Fig. 11.

In Table I, we can find that the accuracy of anchor-based methods is generally higher than that of anchor-free methods. Our method achieves 75.74% on mAP. The detection results illustrate that our method achieves favorable performance against anchor-free methods. Compared with anchor-based methods, our method performs comparably with most of them in accuracy.

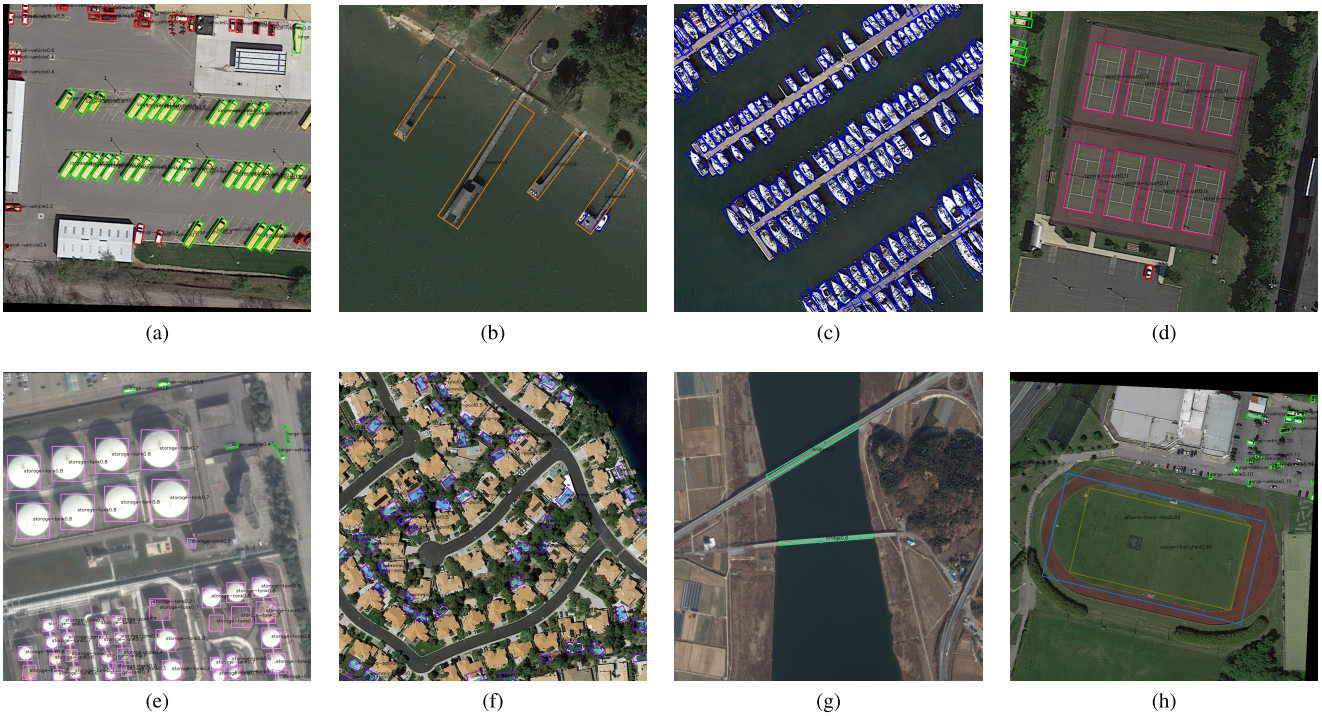
2) Results on HRSC2016 Data Set: Table II shows the comparison results with state-of-the-art methods on the HRSC2016 OBB task. We plot the comparison of accuracy versus speed among different methods in Fig. 10. Our method achieves 90.5% on mAP. Our method is lower than 92.7% of DRN [28], due to DRN that uses the Hourglass-104 as a backbone network and runs at a slower speed of 5.7 FPS. Our method uses DLA-34 as the backbone network and the

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TABLE II

QUANTITATIVE COMPARISONS BETWEEN THE OUR METHOD AND OTHER METHODS FOR OBB TASK ON HRSC2016 TEST SET

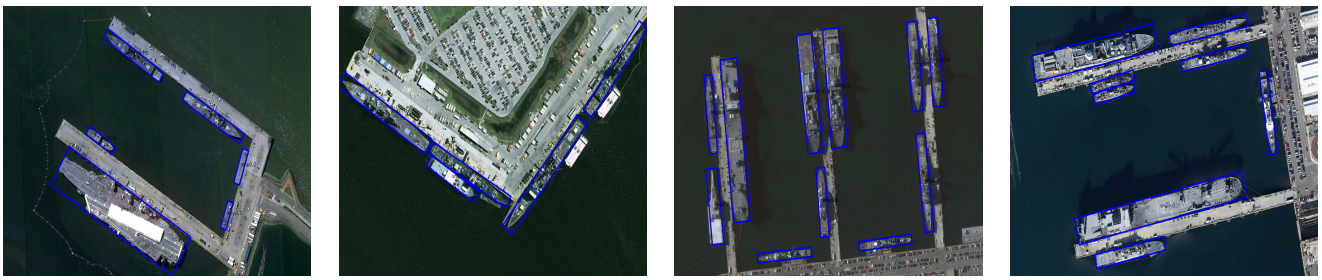
Method	R^2CNN [61]	RRPN [31]	R2PN [62]	RRD [63]	RoI-Transformer [23]	R^3Det [38]	PIou [42]	DRN [28]	Ours
Backbone	ResNet101	ResNet101	VGG16	VGG16	ResNet101	MobileNetV2	DLA-34	Hourglass-104	DLA-34
mAP(%)	73.1	79.1	79.6	84.3	86.2	86.7	89.2	**92.7**	90.5



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 11. Examples of detection results on the DOTA Test Set. (a) Large vehicle and small vehicle. (b) Harbor. (c) Ship. (d) Tennis court. (e) Storage tank. (f) Swimming pool. (g) Bridge. (h) Ground track field and soccer ball field.



(a) (b) (c) (d)

Fig. 12. Examples of detection results on the HRSC2016 test set. Ships in (a) different aspect ratios, (b) different orientations, (c) densely arranged, and (d) different sizes.

detection speed is about 50 FPS. At the same time, our method is 1.3% higher than the PIoU [42] method, which also applies DLA-34 as the backbone network. Among these compared approaches, our method achieves a balance between speed and accuracy. Besides, we present some visualization results on the HRSC2016 test set in Fig. 12.

3) Analysis of CBDA Module: The CBDA module contains center region attention and boundary region attention. To explore the role of each part of attention, we do experiments using different attention components. In the experiments, we add a rotation angle branch to the original CenterNet network structure as a baseline. At the same time, we separately add only center attention, only boundary attention, and CBDA based on the baseline for experiments. We adopt the DLA-34 model as a backbone network and train the whole model on the DOTA training set. We test on the DOTA validation set and present the experimental results in Table III.

TABLE III

COMPARISON WITH DIFFERENT ATTENTIONS ON DOTA VALIDATION SET

Method	+Traditional mask attention	+Center region attention	+Boundary region attention	mAP(%)
Baseline				66.42
Our methods	✓			67.45
			✓	68.24
		✓		68.73
		✓	✓	**69.06**

It can be seen from Table III that when we only use the center region attention, the mAP is 2.31% higher than the baseline model. When we only use the boundary region

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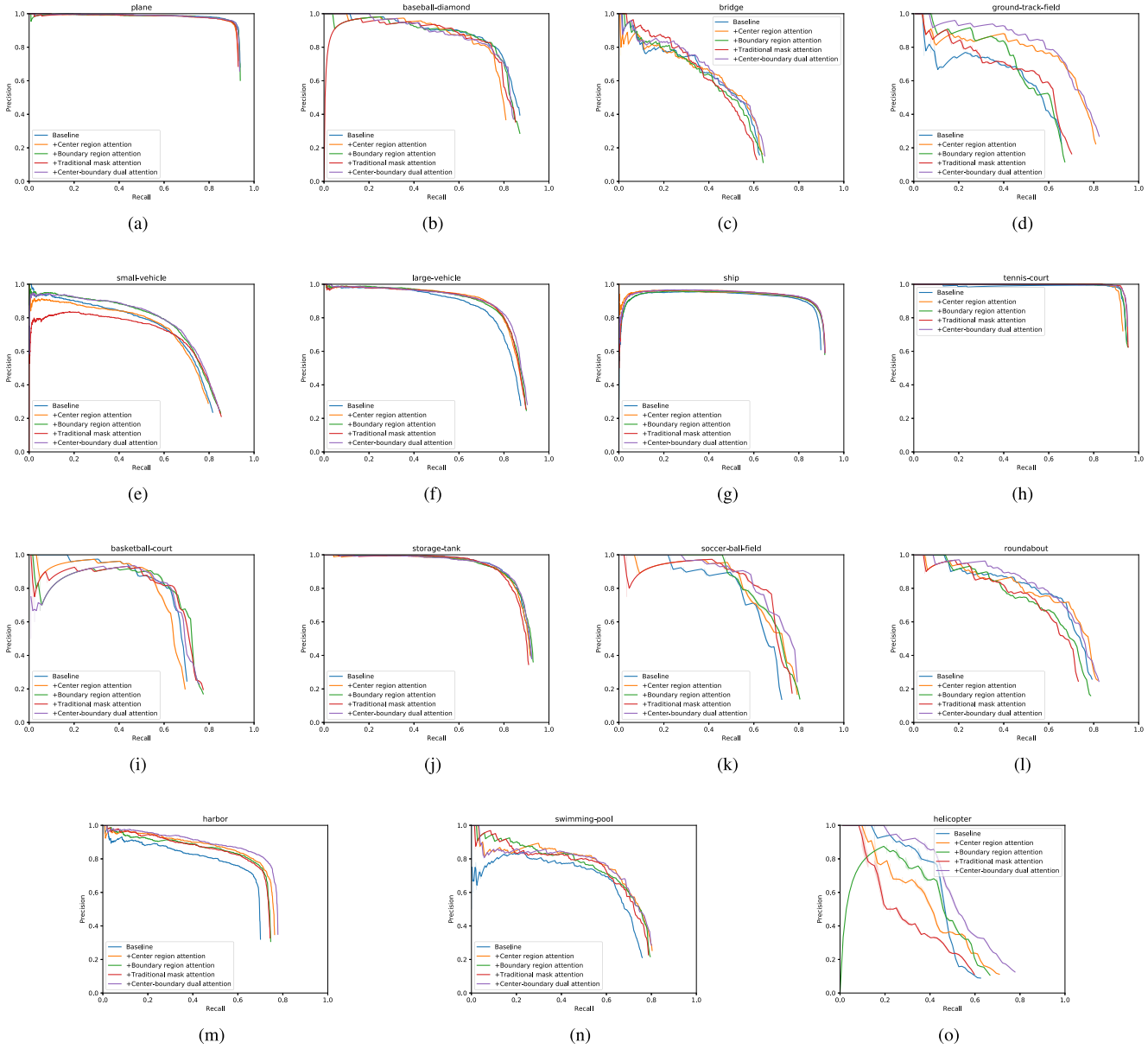


Fig. 13. PR curves of different categories in the DOTA Validation Set. Each form of attention is helpful to the improvement of detection effect, and the effect of our center-boundary dual region attention is generally better than other forms of attention. Better viewed in color with zoomed-in view. PR curves of (a) plane, (b) baseball diamond, (c) bridge, (d) ground field track, (e) small vehicle, (f) large vehicle, (g) ship, (h) tennis court, (i) basketball court, (j) storage tank, (k) soccer ball field, (l) roundabout, (m) harbor, (n) swimming pool, and (o) helicopter.

attention, the mAP is 1.31% higher than the baseline model. When we use CBDA, the mAP can reach up to 69.06%, which is 3.44% higher than the baseline model.

To further prove the effectiveness of our region attention, we use traditional masks generated by polygon vertices for supervised learning and compare the results with our method in Table III. As shown in Table III, using traditional mask for attention supervision can achieve 67.45% on mAP, which is higher than the baseline model, but it is still lower than our proposed CBDA module.

To study the specific effect of different attention modules on object detection, we draw PR curves using different attention methods on the DOTA validation set. It can be seen from Fig. 13 that in most categories, it is beneficial to use the attention module for improving the accuracy and recall of oriented object detection, which indicates that using attention mechanism can distinguish the objects from complex background and extract effective features of oriented objects. However, we also notice that the use of attention mechanism fails in some categories, and the detection effect is not as good as the baseline model.

Specifically, using traditional mask attention will improve the recall rate of small objects such as small vehicle, but it also sacrifices the detection accuracy. Besides, the use of mask attention for roundabout will damage the detection results. Because the image characteristics of objects such as roundabout are more complicated, using the entire bounding box as the mask would bring noisy information from the background.

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TABLE IV

MODEL SENSITIVITY ON DIFFERENT TRAINING AND TESTING SPLITS. THE RESULTS INDICATE MAP ON THE HRSC DATA SET

Methods	Exp1	Exp2	Exp3	Exp4	Exp5	Exp6	Exp7	Exp8	Exp9	Exp10	Mean	Std	P
RetinaNet	56.21	58.27	58.15	51.92	55.14	59.62	50.23	56.26	54.89	56.13	55.68	2.72	3.21e-9
Faster-RCNN	75.41	74.86	76.11	73.81	75.66	75.44	71.03	73.77	74.64	76.02	74.68	1.44	2.06e-17
Ours	80.13	79.34	80.77	80.28	79.23	78.83	82.97	79.83	79.98	79.67	80.16	1.11	*

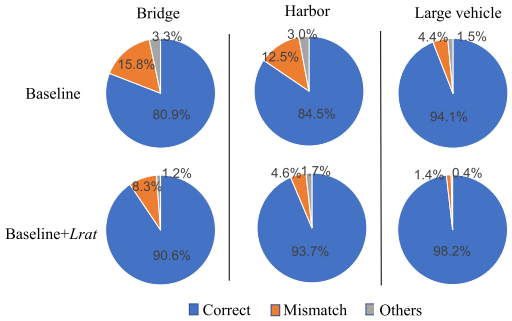


Fig. 14. Visualization of angle improvement. Different pie charts illustrate the percentage of different IoU value types.

TABLE V

ABLATION EXPERIMENTS RESULTS ON THE DOTA VALIDATION SET

Method		CBDA	arwLoss	mAP(%)
Resdcn-18	Baseline			51.02
	Our method	✓		61.42
			✓	61.88
		✓	✓	**62.73**
DLA-34	Baseline			66.42
	Our method	✓		69.06
			✓	69.65
		✓	✓	**70.07**

TABLE VI

PARAMETERS OF CBDA MODULE ON THE DOTA DATA SET

CBDA module architecture	CBDA module layer	Nume of parameters
Conv-center-feature	Conv1	$64 \times (1 \times 1 \times 64) = 4096$
	BN	2
	ReLU	0
Conv-boundary-feature	Conv1	$64 \times (1 \times 1 \times 64) = 4096$
	BN	2
	ReLU	0
Conv-center	Conv3	$256 \times (3 \times 3 \times 64 + 1) = 147712$
	ReLU	0
	Conv1	$15 * (1 \times 1 \times 256 + 1) = 3855$
Conv-boundary	Conv3	$256 \times (3 \times 3 \times 64 + 1) = 147712$
	ReLU	0
	Conv1	$15 * (1 \times 1 \times 256 + 1) = 3855$
Center-attention	Conv1	$64 \times (1 \times 1 \times 64) = 4096$
	ReLU	0
Boundary-attention	Conv1	$64 \times (1 \times 1 \times 64) = 4096$
	ReLU	0
Selective operation	Sigmoid	0
Total	*	319522

Using center region attention can also improve the detection accuracy of slender objects, such as harbor and bridge. At the same time, it is worth noting that the detection accuracy of objects with obvious center region characteristics is better, like the swimming pool. However, it is not effective in the categories where the center region and the boundary region are not obvious, such as basketball court and tennis court, thus only paying attention to the center region features, which is not conducive to object detection. Similarly, the detection accuracy of using only boundary region attention is better than the baseline, but it is worse for some categories whose features are not on the boundary, such as roundabout and helicopter.

When using CBDA, the object detection accuracy and recall in most categories have improved. In object categories that are hard to distinguish from background, the object detection accuracy has also been significantly improved, such as ground track filed and soccer ball filed. At the same time, for some slender objects, such as harbor, bridge, and ship, using the CBDA module can achieve better performance than using only center region attention or boundary region attention.

Experiments results show that it is necessary to focus on different regions for objects in different categories. Overall, each form of attention is helpful to the improvement of detection effect, and the effect of using CBDA is generally better than other forms of attention.

4) Analysis of arwLoss: To study the rotation angle improvement brought by arwLoss, we analyze the percentages of different IoUs with the ground-truth boxes on the DOTA validation set. In the experiment, we select three representative categories: ship, harbor, and small vehicle. These categories are generally relatively slender objects and the angle errors have a greater impact on their IoU calculation.

In the calculation of IoU with ground-truth box, we replace the predicted heatmap and size with the true value of the heatmap and the true value of size and only observe the impact of the predicted rotation angle on the oriented object detection result. The results are shown in Fig. 14, where different colors indicate different IoU value types.

1) *Correct:* The category is correctly detected, and the IoU between the bounding box and the ground truth is greater than 0.5.
2) *Mismatch:* The category is correctly detected, but the IoU between the bounding box and the ground truth is between 0.1 and 0.5.
3) *Others:* The IoU between the bounding box and the ground truth is below 0.1.

It can be seen from Fig. 14 that after using the arwLoss, the proportion of mismatch significantly drops. Specifically, in the bridge category, the proportion of mismatch dropped from 15.8% to 8.3%, whereas the proportion of correct rose from 80.9% to 90.6%. In the harbor category, the proportion of mismatch dropped by 7.9% and the proportion of correct increased by 9.2%. In the large-vehicle category, the proportion of mismatch decreased by 3%, and the proportion of correct increased by 4.1%. Besides, the proportion of extremely abnormal IoU (IoU < 0.1) caused by angle error has decreased on bridge, harbor, and large vehicle by 2.1%, 1.3%, and 1.1%, respectively. Experiments show that using arwLoss can improve the detection accuracy of slender objects and

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reduce the errors caused by mismatch, especially by rotation angle errors.

C. Statistical Analysis

We use the HRSC data set to analyze the sensitivity of CBDA-Net on different training and testing splits. Another two methods (Faster RCNN and RetinaNet) are used for sufficient comparison. We combine the training and testing splits of HRSC data set and randomly divided them by a ratio of 70%:30% for training and testing, respectively. This experiment is repeated by ten times for CBDA-Net and other competitors, while in each of which the images are randomized. Table IV shows the mAP of three methods on ten experiments. The results of mean value and std demonstrate that our method is less sensitive to data randomization.

Besides, we use the T-test to measure the difference between our method and the other two classical methods. We can observe that $P < 0.05$, which demonstrates that CBDA-Net significantly outperforms companion methods.

D. Ablation Study

We perform a series of ablation experiments on the DOTA validation set. In order to save computation cost and time, we use DLA-34 and Resdcn-18 as a backbone network and test the results on the DOTA verification set. We add a rotation angle branch on the Resdcn-18 model and DLA-34 model as a baseline. We use the same parameter settings and training strategy to ensure a fair comparison. Table V shows the experimental results of different methods.

From Table V, we can see that when the baseline uses DLA-34 as backbone network, baseline + CBDA and baseline + arwloss achieve 2.64% and 3.23% improvement on mAP, respectively. When we use the CBDA module and arwLoss at the same time, the mAP reaches 70.07%, which is 3.65% higher than the baseline. Similarly, it has the same effect on the Resdcn-18 backbone network, and baseline + CBDA and baeline + arwloss achieve 10.4% and 10.86% improvement on mAP, respectively. When we use the CBDA module and arwLoss at the same time, the mAP reaches 62.73%, which is 11.71% higher than the baseline.

Experiments results show that the CBDA module and arwLoss both contribute to the accuracy of the oriented object detection task, and the effect of using both the CBDA module and arwLoss at the same time is better. In addition, our method has the same improvement effect under different backbone networks, which proves the robustness of our method.

E. Complexity Analyzes

For remote sensing image processing, we hope to improve the detection accuracy without sacrificing too much memory loss. In CBDA-Net, we propose the CBDA module and arwLoss to enhance detection accuracy. Meanwhile, CBDA-Net has increased some memory compared with the original network, which mainly comes from the CBDA module. We analyze the complexity of CBDA module and report the parameters of CBDA module in Table VI.

As shown in Table VI, the CBDA module has about 319 522 parameters on the DOTA data set in total, which take only about 1.22 MB of memory. Besides, we test the inference time of the CBDA-Net and baseline. Compared with baseline, the network inference time of our CBDA-Net requires extra 0.003 s on one image, which are pretty small negligible.

V. Conclusion

This article proposes a novel anchor-free oriented object detection method, CBDA-Net, for remote sensing images. In our CBDA-Net, the CBDA module integrates boundary region attention with center region attention, which could accurately extract the features of oriented objects while reducing background interference. Moreover, the proposed arwLoss considers the influence of the object's aspect ratio on angle error, which makes the angle regression of slender objects more accurate. We conduct ablation experiments on the DOTA and HRSC2016 data sets. The results demonstrate that our method achieves better performance against other anchor-free methods with a real-time speed of 50 FPS.

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